Vinamra Mishra

+91-9173255769 | Bengaluru, India | mishravinamra5@gmail.com | GitHub | LinkedIn

Summary

Software Engineer with 2+ years of experience building Java and Rust services across payment identity systems and serverless platforms. Hands-on with zero-downtime service migrations, Kafka event pipelines, and latency optimization on systems serving 5M+ users.

Education

Birla Institute of Technology, Mesra — Ranchi, India
B.Tech in Information Technology (8.80 CGPA) — *November 2020 - May 2024*

Work Experience

Visa (Cybersource) — Bengaluru, India
Software Engineer — *June 2025 - Present*

- Migrated user-management APIs and React portal flows to a Java 17/Vert.x microservice across 5M+ merchant accounts, adding request header signature checks and automatic fallback to the legacy service for zero-downtime rollout.
- Built a Kafka pipeline to sync state changes back to the legacy database with zero data loss, holding partition offsets across 3 retries before routing failed events to a dead-letter queue with operator replay.
- Implemented SAML SSO and directory-group provisioning across 100+ multi-tenant roles, and shipped a zero-downtime SHA-256 to SHA-512 password hashing upgrade at login.

Openreach (BT Group) — Bengaluru, India
Software Development Engineer — *August 2024 - June 2025*

- Migrated an order-processing service from Spring JPA to Quarkus with stored procedures, cutting cold-start time by 60% for serverless deployment.
- Optimized JDBC and database calls, decreasing network roundtrips and cutting response time by 40%.
- Reduced serverless costs by 45% through efficient resource utilization and optimized stored-procedure execution.

Openreach (BT Group) — Bengaluru, India
Software Engineer Intern — *January 2024 - July 2024*

- Migrated a product-availability service to Quarkus 3.8, reducing startup time by 40% and enabling AWS Lambda.
- Implemented resilient SOAP/ESB integration, sustaining 99.9% uptime across 1,000 daily availability checks.

Projects

InScope | *Rust, WebAssembly, Cloudflare Workers, D1 SQLite, Turnstile* — *October 2026*

- Built a serverless LLM chat interface in Rust and WebAssembly on Cloudflare Workers that runs a classifier model before the main LLM call to filter off-topic prompts and sanitizes inputs to block prompt injection.
- Added login-free bot protection using Cloudflare Turnstile, Tor and datacenter IP filtering, HMAC-SHA256 signed bearer tokens, and browser fingerprint scoring across timezone, language, and HTTP protocol.
- Implemented atomic daily rate limiting in Cloudflare D1 SQLite using a single batch query to enforce tiered per-device and global quotas without burning global budget on rejected requests.

Open Source Contributions

- **Arroyo** (April 2025): Added string-slice argument support to Rust user-defined aggregate functions.
- **Zen Browser** (December 2024): Removed the MSYS2/WSL dependency from Windows builds, enabling native builds.

Technical Skills

Languages: Java 17, Rust, SQL, JavaScript
Backend & APIs: Vert.x, Spring Boot, Quarkus, Kafka, REST APIs, Microservices, WebAssembly
Data: Oracle, Microsoft SQL Server, SQLite, Redis
Cloud & DevOps: GCP, AWS Lambda, Cloudflare Workers, Docker, Kubernetes, Helm, Argo CD, Jenkins, Git
Testing & Security: JUnit, Mockito, OAuth 2.0, SAML SSO, HMAC, Cloudflare Turnstile
Core CS: Distributed Systems, Data Structures & Algorithms, Operating Systems, DBMS

Achievements

Codeforces | vinam5: Max rating 1450+; global rank 1276 in Round 780 Div 3 amongst 22,000+ participants.
LeetCode | mishravinamra5: 571 problems solved, including 138 hard; contest rating 1,795.